

Dominic Le

Contact

+1 (778) 887-6799

dominic.le@mail.utoronto.ca

[GitHub](#), [Webpage](#)

Research Interests

Causation, counterfactuals, cognitive development, computational cognitive science, learning, memory, philosophy of language, philosophy of cognitive science.

Education

M.A. Developmental Psychology and Education.
University of Toronto

2026 - 2027

H.B.Sc. Psychology and Cognitive Science, *High Distinction*.
University of Toronto

2022 - 2026

Ongoing Research

Backwards Counterfactual Reasoning in Children and Adults
Supervisor: Ioana Grosu & Patricia Ganea, University of Toronto.
Department of Applied Psychology and Human Development.

Sep 2024 –

Hierarchical Models of Associative Learning
Supervisor: Sara Aronowitz, University of Toronto.
Department of Philosophy.

Aug 2024 –

Modelling Causal Reasoning in a Cognitive Architecture
Supervisor: Can Mekik, University of Toronto.
Department of Psychology.

Sep 2024 –

Peer-reviewed Publications

Proceedings

Dominic Le, Can Mekik (2026). *A Spreading Activation Model of Causal Reasoning*. Proceedings of the 23rd International Conference on Cognitive Modelling. Forthcoming.

Ioana Grosu, **Dominic Le**, Patricia Ganea (2025). *Backwards Counterfactuals and the Closest Possible World*. Proceedings of the Annual Meeting of the Cognitive Science Society. Vol. 47. 2025. <https://escholarship.org/uc/item/0q93p7sc>

Conferences & Presentations

Refereed

Dominic Le, Can Mekik (2026). *A Spreading Activation Model of Causal Reasoning*. Talk to be presented at the International Computational Cognitive Modelling Conference (ICCM, July), Montreal, CA. Poster presented at the Canadian Society for Brain, Behaviour and Cognitive Science Annual Meeting (CSBBCS, June), North York, CA.

Dominic Le, Ioana Grosu, & Patricia Ganea (2026, April). *Children's Backward Counterfactual Logic*. Poster presented at the Cognitive Development Society Bi-Ennial Conference (CDS), Montreal, CA.

Dominic Le (2025). *Causal Stacks: A Structural Framework for Recurrent and Hierarchical Counterfactual Reasoning*. Presented as poster for the 5th European Experimental Philosophy Conference (EXPhi, July), Norwich, UK; the Society for Philosophy and Psychology (SPP, June) Annual Meeting; and the Annual Meeting of the Cognitive Science Society (CogSci, August).

Ioana Grosu, **Dominic Le**, & Patricia Ganea (2025, July). *Backwards Counterfactuals and the Closest Possible World*. Talk presented at CogSci 2025, San Francisco, US.

Ioana Grosu, **Dominic Le**, & Patricia Ganea (2025, June). *Changes to Models of Reality in Backwards Counterfactual Reasoning*. Poster presented at the 51st Annual Meeting of the Society for Philosophy and Psychology (SPP), Ithaca, US.

Student

Dominic Le, Can Mekik (2026, April). *How Search Strategies Explain Directional Asymmetries in Causal Reasoning*. Talk presented at the Psychology Undergraduate Research Community (PURC) Student Symposium, Toronto, CA.

Dominic Le, Ioana Grosu, & Patricia Ganea (2025, February). *Time's Asymmetry in Counterfactuals*. Talk presented at the Cognitive Science and Artificial Intelligence Student Association (CASA) Undergraduate Conference, Toronto, CA.

Dominic Le (2024, September). *Modeling Children's Development of Modal Thought*. Talk presented at the Collective Undergraduate Research (CURE) Flash Talks, Toronto, CA.

Dominic Le, Oksana Tkachman (2024, March). *Spatial Frames of Reference in Signed Languages*. Talk presented at the Western Interdisciplinary Student Symposium on Language Research (WISLR), London, CA.

Software	Quentin Dercon, Rina Harsch, Dominic Le , Shannon Yasuda, Cherrie Chang, Melissa Struhl (2025). <i>snap-sort (Version 1.0.0)</i> [Software]. In review for jsPsych-contrib. https://github.com/DomDaSquishyBomb/plugin-drag-and-drop	
Labwork	Research Assistant, Cognitive Development	Jan 2024 –
	PI: Ioana Grosu & Patricia Ganea, University of Toronto Language & Learning Lab	
	Research Assistant, Psycholinguistics	Mar 2023 –Sep 2024
	PI: Oksana Tkachman, University of British Columbia Language & Learning Lab	
	Research Assistant, Morality and Abnormal Psychology	Nov 2023 –Jun 2024
	PI: Mark Miller, University of Toronto Consciousness & Wisdom Studies Lab	
Previous Employment	Child's Program Assistant , UofT Family Housing	Jun 2024 –Aug 2024
	Supported Child Care Worker , BGC Canada	Jun 2023 –Sep 2023
	Day Camp Coordinator , Cadets Canada	Jun 2022 –Aug 2022
Relevant Undergrad Coursework		

	<u>Programming & Computer Science:</u> Data, Computation, and the Mind: Python Software Design: Java <u>Psychology & Cognitive Science:</u> Behavioural Neuroscience Sensation and Perception Intro to Philosophy: Knowledge and Reality Intro to Developmental Psychology <u>Statistics & Mathematics:</u> Experimental Design and Theory: R Psychology Statistics II: JASP Probability and Statistics I: R Linear Algebra	Seminar on Cognitive Architectures: Python Intro to A.I.: Python Intro to Computer Programming: Python Minds and Machines Semantic Theory Intro to Cognitive Science Intro to Psychology Intro to Cognitive Psychology Treatment of Psychological Data: R Math Expres. and Reas. in Computer Science Probability and Statistics II: R Multivariable Calculus with Proofs Calculus with Proofs
Student Affiliations	Editor , Mentis - UofT Cognitive Science Undergraduate Journal Mentor , CURE - Undergraduate Research Student Group Staff Writer , Demo Magazine - Hart House Music Publication Executive Secretary , CASA - CogSci and A.I. Student Association CogSci Course Union Representative , University College Council	2024 – 2024 – 2023 - 2024 2024 –2025 2024 –2025
Awards & Scholarships	(\$7500) University of Toronto Research Excellence Award (UTEA-SSH). University of Toronto Student Leadership Award. (\$7500) University of Toronto Research Excellence Award (UTEA-SSH). (\$1500) Victoria College Regents Participation Award. (Priceless) MAT237 Meme Contest Nominee. x2 (\$2000) Gyro Club of Nanaimo Jack Ryan Scholarship. (\$1250) British Columbia District/Authority Scholarship. (\$1250) British Columbia Excellence Scholarship. (\$2000) Vancouver Island University Dual Enrollment Scholarship.	2026 2026 2025 2025 2024 2022 2022 2022 2021
Travel Awards	(\$500) Dr. Roseann Runte Award. (CDS) (\$1000) Susan Macdonald Award. (CogSci) (\$250) ASSU Travel Grant. (EXPhi) (\$150) SPP Travel Award.	2026 2025 2025 2025
Skills & Characteristics	<u>Natural Languages:</u> <u>Computer Languages:</u> <u>Certifications:</u> <u>Software Tools:</u> <u>Outside Interests:</u>	Fluent: English (native), French. Rudimentary: Vietnamese (first-learned). Proficient: R, Python, Java, HTML/CSS Rudimentary: JavaScript, C#, C++. Standard First Aid - with CPR-C-AED (St. John Ambulance till 2026), Responsible Adult (YMCA, 2023) Document-preparation: MS Office, LaTeX (Overleaf/VSCode), Google Workspace. Data-collection: Qualtrics, MTurk, ChildrenHelpingScience, REDCap, jsPsych. Data-processing: RStudio, Pandas+NumPy, JASP. Image-manipulation: Photoshop, Canva Audio-editing: Ableton Live, Logic, Audacity. Video-editing: Premiere, Final Cut, Animate. Songwriting and role-playing video games.